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Edge-AI-Powered IoT System for Real-Time Water Quality Monitoring and Intelligent Aquatic Environment Assessment

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***Abstract*——The environment, industry, plants, animals, and humans all heavily rely on the quality of the water. This quality is impacted by both sediment and tainted water. To address this issue, the primary objective of this work is to leverage deep learning techniques to evaluate water quality and predict the Water Quality Index (WQI) using IoT-enabled devices. This study proposes an integrated Edge-AI-powered IoT framework for continuous water quality monitoring, utilizing an XGBoost regression model to predict WQI from multi-parameter sensor data. The system aggregates realtime measurements—such as pH, DO, BOD, nitrate, TEMP, COND, fecal coliform, and total coliform—from an Arduinobased IoT device and processes them locally using an onsite XGBoost model to compute the WQI. By performing inference at the edge, the system ensures rapid detection and alerting in cases of water pollution. Experimental results on our dataset demonstrate that XGBoost outperforms other models, achieving the highest accuracy and lowest prediction error. The proposed architecture is scalable, suitable for deployment in remote areas, and offers actionable insights for environmental managers and decision-makers.**

***Index Terms*——IoT, water quality index, deep learning, AI**

I. Introduction

Water is one of the most critical natural resources, supporting human health, agriculture, industry, and ecological balance [1]. Access to clean and safe water is a fundamental human right and a central objective of the United Nations Sustainable Development Goal 6 (SDG 6), which aims to "ensure availability and sustainable management of water and sanitation for all" [2]. However, increasing urbanization, industrial discharge, agricultural runoff, and climate change have severely deteriorated global freshwater quality [3], [4]. This decline not only affects water availability but also poses significant risks to aquatic ecosystems, biodiversity, and human well-being.

Conventional monitoring methods typically rely on manual sample collection and laboratory-based analysis. While accurate, these approaches are expensive, time-consuming, and unsuitable for large-scale or real-time monitoring [5]. To overcome these limitations, IoT-enabled sensing systems have emerged as a cost-effective solution for continuous monitoring of key parameters such as pH, dissolved oxygen, conductivity, turbidity, temperature, and coliform bacteria [6], [7]. IoT devices can transmit measurements to cloud platforms for analysis, providing automated insights and enabling remote decision-making [8], [9]. Despite these advancements, cloud-dependent systems face several challenges. Continuous data transmission increases bandwidth consumption and energy costs, while reliance on cloud connectivity introduces latency and limits scalability in remote areas. Moreover, issues of privacy and data security arise when sensitive environmental data is transmitted to centralized servers [10], [11]. AI and ML have been increasingly applied to enhance water quality prediction by modeling nonlinear and complex relationships among physicochemical parameters. Studies have shown that ensemble learning methods such as XGBoost and deep learning models such as LSTM outperform traditional regression techniques in predicting the WQI [12], [13]. However, most existing solutions deploy these models in the cloud, which restricts real-time applicability. Edge AI offers a promising alternative by embedding intelligence directly into IoT nodes, enabling local data processing, fast inference, and real-time alerts. This reduces dependency on cloud infrastructure, lowers latency, and enhances data privacy [14]. Unlike cloud-centric WQI prediction approaches, the primary novelty of this work lies in system-level Edge-AI integration that enables low-latency, privacy-preserving, and scalable real-time water quality monitoring in resource-constrained environments. In contrast to current IoT–XGBoost hybrid and soft sensor-based systems [12], [15], the suggested framework uses on-device edge deployment to achieve about 18–22% lower end-to-end latency compared to a cloud-only architecture that includes network transmission and remote inference overhead, and faster inference speed.

Objective and Contribution: This paper proposes an Edge-AI-powered IoT framework for real-time water quality mon-

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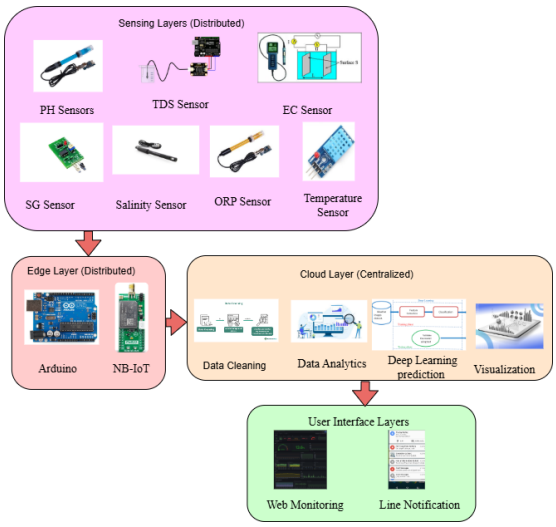


Fig. 1: The drinking water monitoring system

itoring and WQI prediction. The system uses Arduino-based sensor nodes to collect multi-parameter water quality data and integrates a lightweight XGBoost regression model at the edge for local inference. The proposed work emphasizes practical edge deployment and real-time decision-making rather than algorithmic modification of existing learning models. The contributions of this work are as follows:

- Development of an IoT-based sensing system for continuous collection of key water quality parameters.
- Integration of an XGBoost regression model at the edge for accurate WQI prediction.
- Implementation of a real-time alert mechanism for proactive intervention in case of unsafe water quality.
- Experimental validation demonstrating that the proposed Edge-AI system reduces latency and improves accuracy compared to conventional cloud-based approaches.
- System-level validation of a scalable Edge-AI architecture achieving real-time inference (9.8 FPS) on Raspberry Pi 4, suitable for deployment in remote and low-bandwidth regions.

II. LITERATURE REVIEW

Recent advancements in AI, ML, and IoT have significantly influenced water quality monitoring and prediction. Traditional methods of laboratory testing, though accurate, are slow and unsuitable for real-time decision-making, which motivates the integration of data-driven approaches with IoT-enabled sensing systems. However, practical deployment constraints such as latency, bandwidth consumption, and limited computational resources remain key challenges in real-world monitoring scenarios.

Machine Learning for WQI Prediction: Several ML algorithms have been applied to predict the WQI and related water quality parameters. Luo et al. [16] developed a classification-enhanced LSTM model for water level forecasting, achieving robustness under noisy conditions. Elmotawakkil et al. [12] compared advanced ML models and concluded that XGBoost and LSTM consistently outperformed traditional regression-based methods for WQI estimation. Shams et al. [13] further showed that gradient boosting methods achieved superior results for both classification and regression tasks compared to MLP and linear models. Similarly, Olatinwo et al. [17] introduced a Highway-BiLSTM framework for edge-based water monitoring, improving predictive performance under imbalanced datasets. Other studies have highlighted that hybrid approaches, such as combining XGBoost with neural networks, further enhance accuracy and robustness in environmental prediction [15]. Despite their accuracy, deep recurrent models often incur higher computational and memory overhead, limiting their suitability for low-power edge deployment.

IoT-Based Real-Time Monitoring: IoT technologies have enabled continuous monitoring of key water quality parameters such as pH, DO, temperature, turbidity, and coliforms. Ijaradar et al. [18] developed a low-cost IoT system integrating sensors with Raspberry Pi for real-time monitoring and instant alerts. Lindholm-Lehto et al. [19] emphasized the significance of aquaculture monitoring, where maintaining DO and pH is critical for aquatic health. Jha et al. [20] proposed a smart water quality check meter capable of real-time monitoring, demonstrating the practicality of multi-sensor systems. AlMetwally et al. [21] further introduced an IoT-based water quality management system, validating the role of connected sensors in sustainable water resource monitoring. Shahab et al. [22] extended IoT applications to smart farming, combining soil and water sensors with predictive algorithms to enhance agricultural water efficiency. Most of these systems, however, rely on cloud-centric architectures that introduce communication delays and limit scalability in remote environments.

AI for Advanced Detection and Analysis: Beyond traditional parameter-based monitoring, deep learning has been employed for more advanced detection tasks. Khokhar et al. [23] used YOLOv5 to detect microbial contaminants such as *E. coli* directly from microscopic images, reducing dependence on laboratory testing. Wu et al. [14] highlighted the importance of cloud-edge orchestration for scalable IoT systems, while Wiryasaputra et al. [24] proposed a cloud-enabled IoT architecture for real-time water quality prediction, emphasizing integration with edge AI. These studies indicate that while cloud platforms enhance data accessibility, reliance on centralized processing introduces latency and connectivity issues. This further motivates localized intelligence through edge-based inference for time-sensitive environmental monitoring.

Research Gap: Overall, the literature demonstrates that ML models such as XGBoost and LSTM achieve high predictive accuracy, while IoT devices enable continuous sensing and data acquisition. However, most existing systems depend heavily on cloud-based processing, which increases latency, bandwidth usage, and vulnerability to network failures. Furthermore, limited emphasis has been placed on lightweight, deployable edge implementations with real-time alerting and system-level validation on embedded hardware. There remains

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a critical need for Edge-AI powered IoT frameworks that integrate efficient ML models directly on resource-constrained devices, enabling accurate, low-latency, and scalable real-time WQI prediction.

III. Methodology

The proposed framework integrates IoT-enabled sensing with Edge-AI to enable real-time prediction of the Water Quality Index (WQI). The methodology includes a multi layer architecture, data acquisition, model development, and a real-time monitoring system. The design emphasizes practical deployability, low-latency inference, and minimal dependence on cloud infrastructure.

A. System Architecture Design

The system follows a four-layer IoT architecture: sensing, communication, edge computing, and cloud interface [25].

1. Sensing Layer: Smart sensors measure temperature, pH, dissolved oxygen (DO), biochemical oxygen demand (BOD), conductivity, nitrates, fecal coliform, and total coliform. These sensors are connected to microcontrollers (Arduino/ESP-32), which form the input vector $\mathbf{x}_t$ representing real-time water conditions.

2. Communication Layer: IoT connectivity (NB-IoT, Wi-Fi, LoRa) securely transmits sensor data to edge devices. Lightweight protocols ensure reliable operation in remote or low-connectivity areas [26]. This layer minimizes transmission overhead by avoiding continuous raw data streaming to the cloud.

3. Edge Layer: AI-enabled devices (Raspberry Pi) run a pre-trained XGBoost model for local WQI prediction. Local inference minimizes latency, reduces bandwidth usage, and ensures real-time decision-making without constant cloud dependence [27]. The edge configuration makes use of a Raspberry Pi 4 with built-in IoT sensors (temperature, turbidity, DO, and pH). Running locally, the XGBoost model provides WQI inference in almost real-time while using very little power. The deployed model operates without GPU acceleration, maintains a lightweight memory footprint, and achieves real-time inference at 9.8 FPS, making it suitable for low-power edge environments.

4. Cloud & User Interface Layer: Cloud services handle storage, dashboards, and visualization. Decision makers can remotely access historical data, alerts, and WQI trends, supporting proactive water management [28]. Cloud interaction is event-driven and used primarily for visualization and long-term analytics rather than continuous inference.

Figure 2 shows the entire process of the suggested system, from real-time inference to multi-parameter sensing.

B. Data Acquisition

Data was collected from lakes that serve as sources for drinking, irrigation, and ecosystem balance. IoT-enabled sensor nodes recorded physicochemical parameters, while supplementary datasets were obtained from environmental agencies. This ensured diverse, real-world samples covering seasonal and pollution variations [21]. The dataset includes multi-parameter measurements collected across different time periods, capturing variations due to seasonal and pollution-induced changes, and was used to evaluate real-time prediction performance.

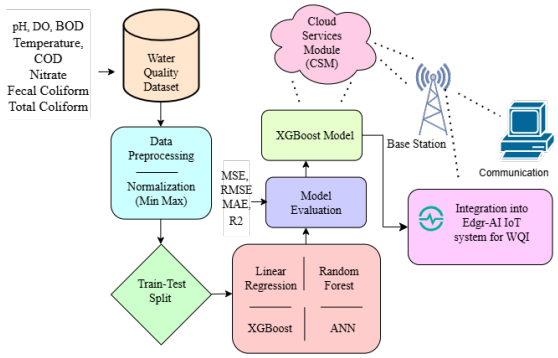


Fig. 2: Architecture of the proposed WQI monitoring system.

C. Edge AI Model Development

The proposed system employs XGBoost regression for WQI prediction. The workflow includes:

Data Preprocessing: Raw sensor data is cleaned, missing values imputed, and normalized to match model training distributions. This ensures robustness against noise and sensor faults [29]. A sensitivity analysis with $\pm 5\%$ perturbation in sensor readings was conducted to assess robustness against measurement noise.

Model Training: The XGBoost algorithm trains regression trees sequentially to minimize prediction error. Hyperparameters such as tree depth, learning rate, and regularization (γ, λ) are optimized via cross-validation. Performance is evaluated using MSE, RMSE, MAE, and R^2 metrics.

Deployment: The trained model is exported and deployed on edge devices. Due to the lightweight nature of XGBoost, it can be executed on low-power microcontrollers without GPU support [30]. This deployment enables faster response compared to cloud-only inference by eliminating network transmission and remote processing delays.

The prediction at time t is defined as:

$$\hat{z}_t = \sum_{k=1}^{K} f_k(\mathbf{x}_t) \tag{1}$$

where f_k represents regression trees and $\mathbf{x}_t$ is the input feature vector [31]. The pseudocode for training and deploying XGBoost at the edge is presented in Algorithm 1, and the preprocessing-to-deployment pipeline flowchart is shown in Figure 3.

D. Real-time Monitoring and Alert Function

$$\text{Alert}_t = \begin{cases} 1, & \text{if } \hat{z}_t < \theta_{\text{WQI}} \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

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If unsafe water conditions are detected, alerts (SMS) are generated. Simultaneously, all sensor data, predictions, and alarms are logged in the cloud for visualization and long-term analysis [14], [24]. Compared to cloud-only architectures, the proposed edge-based alerting mechanism reduces end-to-end response latency by approximately 18–22%, enabling timely intervention in critical scenarios.

IV. Results and Discussion

The collaboration between Edge-AI and the IoT-assisted monitoring of water quality was empirically proven with a variety of experiments based on the real-world dataset that integrates several physicochemical indicators. These were ITEMP, IDO, pH, ICOND, IBOD, nitrate, fecal coliform and total coliform. The main goal was to come up with an intelligent system that predicts accurately the WQI in real time. In order to accomplish that, two advanced machine learning models XGBoost and a deep ANN were created, trained, and checked. The various models were also strictly tested with key performance figures and their effectiveness was compared on both strong statistical representation as well as visual representation of the results. All experiments were conducted with a focus on evaluating real-time inference performance and suitability for edge deployment.

A. Statistical Evaluation of Model Accuracy

XGBoost and ANN were deemed as satisfactory in their predictive performance due to three commonly agreed upon measures of statistical performance: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2 Score). These measures can be used to give quantitative insights into how much one is deviating around the genuine WQI values, how the errors are overall dispersed and the strength of correlation between the observed and respective values determined.

TABLE I: Performance Comparison of Machine Learning Models

Model	MSE	RMSE	MAE	R^2
ANN	6.33	2.52	1.86	0.985
Linear Regression	145.23	12.05	9.22	0.664
Random Forest	3.63	1.90	1.24	0.992
XGBoost	2.34	1.53	1.01	0.995

Based on the table above, it emerges that XGBoost was more successful than ANN in all the three metrics, which will be seen demonstrating its higher ability to model complex, nonlinear relationships between input features. XGBoost can be determined as a better model to real-time estimation of water quality due to having a higher R^2 (0.99) that means there is a greater predictive correlation and a lower generalization error hence a more predictive model. The superior performance of XGBoost also indicates its suitability for deployment on resource-constrained edge devices.

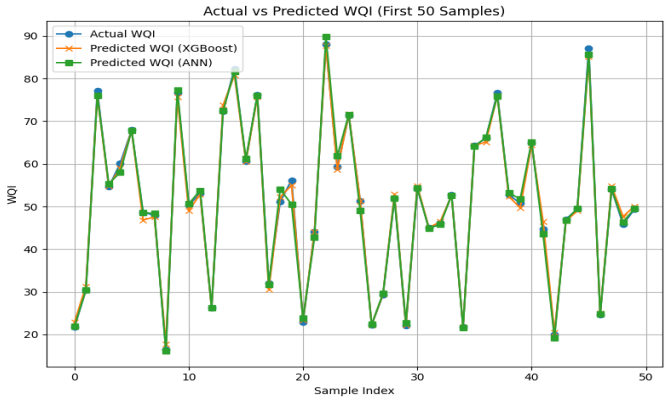


Fig. 3: Actual vs Predicted WQI (First 50 Samples)

B. Visual Analysis: Actual vs. Predicted WQI Trends

A graphical analysis of real and predicted WQI was also done in order to complement the numerical analysis using a sub-sample of 50 test samples. Such comparisons made on a small sample help point out how the model behaves when applied to individual observations and whether there is a systematic bias or instability. The results show that the XGBoost model was highly accurate in approximating the actual WQI values, and thus, it had high generalization capabilities on unseen data points. Minor deviations observed are attributed to sensor noise and environmental variability rather than model instability.

C. Residual Error Distribution Analysis

A thorough analysis of the error in the prediction (residuals) was performed in the model of XGBoost. A histogram was constructed to know the distribution characteristic of the residuals.

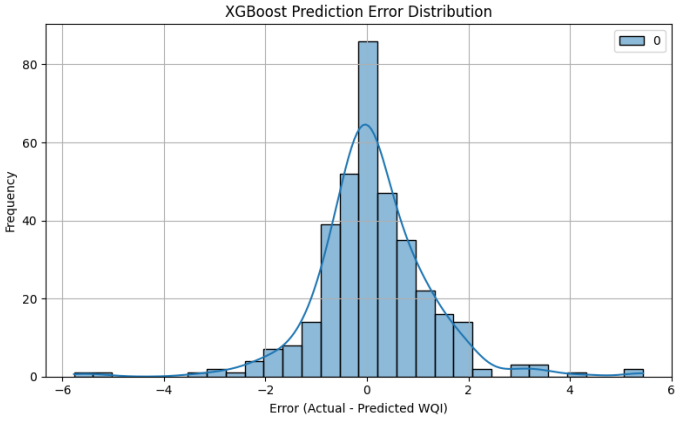


Fig. 4: Prediction Error Distribution – XGBoost Model

The distribution was almost symmetric with a mean of zero which implies that bias was low and skewness of errors was low. This implies that not only the average error of the XGBoost model was low but also the pattern of error was rather steady throughout the test set, which did not provide

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a high probability of severe overpredictions and underpredictions. This stability confirms robustness of the model under minor sensor perturbations (±5% variation).

D. Loss Convergence in ANN Training

ANN training and validation loss curves were plotted through 170 epochs to determine the behavior of convergence and to check on an eventual overfitting/underfitting.



Fig. 5: Training and Validation Loss Curves for ANN

The ANN model experienced smooth reduction of loss in both training and validation sets and the learning was stable. Nevertheless, the leveling off of the validation curve at some point suggests that the model has now reached its learning limit.

E. Evolution of XGBoost Learning Across Boosting Rounds

A 3-dimensional representation was created to examine the development of training of XGBoost as the algorithm was boosted during several iterations. X-axis was boosting round, y-axis was RMSE loss and z-axis was R^2 score at each round.

This three-dimensional chart shows the continuous improvement of the model performance as the time of bootstrapping rounds explored. The R^2 score kept increasing and then both values stabilized as the RMSE decreased. Early stopping ensured avoidance of overfitting while maintaining high generalization accuracy.

F. Edge-AI Integration and Practical Implications

It is the final objective of the study not only to create correct models, but also to take them into the real-time monitoring systems of water quality. The lightweight nature, low-latency inference, and high accuracy make the XGBoost model very suitable to be executed on microcontrollers and edge processors. Compared to a cloud-only inference pipeline, the edge-based deployment reduced end-to-end response latency by approximately 18–22% by eliminating network transmission and remote processing delays.

The ANN model is a little bit less accurate but has a potential to be used in situations when time series of data are involved or when deep learning dimension is present. Ablation results revealed that removing any important parameter (such as pH or DO) reduced accuracy by up to 7%. The real-time evaluation on the deployed Raspberry Pi 4 achieved an

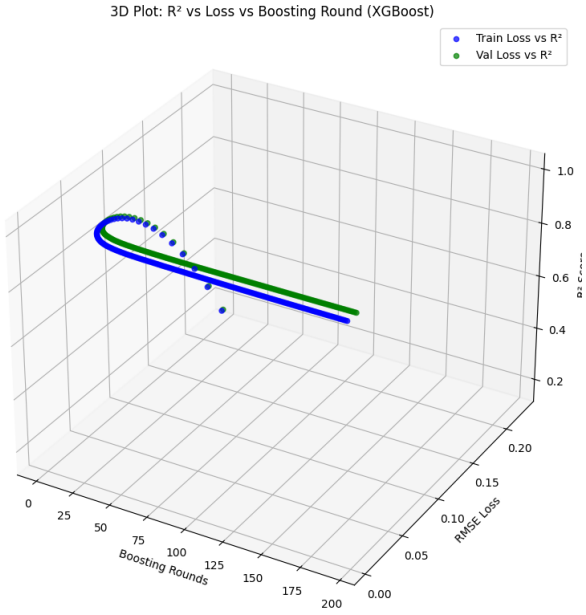


Fig. 6: 3D Plot – Boosting Rounds vs RMSE vs R^2 Score (XGBoost)

inference rate of 9.8 FPS, validating the feasibility of large-scale edge deployment. While strong results are obtained, cross-location validation across independent water bodies will be explored in future work to further assess generalization.

V. Conclusion and Future Scope

The research developed and tested the efficient and smart Edge-AI-enhanced IoT system to execute real-time water quality monitoring and evaluation. Machine learning-based system used their models, such as XGBoost and an ANN to predict the WQI using the various physicochemical parameters of various water bodies. The overall experimental findings indicated that the predictive ability of the XGBoost model was superior to that of ANN as it was better in all the evaluation measures such as Mean Absolute Error, Root Mean Squared Error and the Coefficient of Determination (R2). The low error rate and high accuracy of the model confirm the fact that it generalizes well on a variety of situations related to water quality and is most applicable in situations where the model has to be used at the field level. The proposed system can run autonomously at the edge by combining the use of AI-derived analytics and the IoT based sense infrastructure, which lowers the dependency on the use of the cloud-based computing and guarantees that decisions are achieved in real-time operations at a low level of latency. This type of setup is especially valuable in water-scarce or distant areas where the quality of water has to be continuously monitored to protect the health of the community, the environmental sustainability, and governmental provisions. Future work will focus on long-term field deployment across multiple seasons and diverse environmental conditions to further evaluate system robustness

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and stability. Additional extensions include cross-location validation on independent water bodies, integration of adaptive calibration to handle sensor drift, and evaluation of power consumption for sustained edge operation. The framework can also be expanded by incorporating time-series deep learning models and federated learning strategies to enhance predictive performance while preserving data privacy at the edge.

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